

Getting Started

Laptop/bwUniCluster 3.0

Julia on Your Laptop

Prerequisites

Course materials

<https://github.com/carstenbauer/JuliaKIT26>

Julia

`julia` (version 1.12)

Jupyter + VS Code

`jupyter lab`
`code`

Julia VS Code integration via extension.

The screenshot displays the VS Code interface with the Julia extension installed. The Explorer sidebar on the left shows the project structure, including a `JULIAINTERPRETER` section with files like `.github`, `.vscode`, `benchmark`, `bin`, `deps`, `docs`, `src`, `test`, `.gitignore`, `.travis.yml`, `appveyor.yml`, and `LICENSE.md`. Below this is the `JULIA EXPLORER: JULIA WORKSPACE` section, which lists the workspace structure, including `Core`, `InteractiveUtils`, `PlotThemes`, `Plots`, and `ans`. The `ans` section shows the current state of the REPL, including the `backend` (Plots.GRBackend()), `n` (1), `attr` (DefaultsDict{Symbol, A...}), `series_list` (Array{Plots.Seri...}), and `nothing`.

The main editor area shows a Julia script in `Untitled-1`. The code defines a function `mandel` and uses the `map` function to iterate over a range. The code is as follows:

```
9      ...end
10     ...return maxiter
11 end mandel (generic function with 1 method)
12
13 for i in 1:10
14     ...println(i)
15 end
16
17 map
```

A tooltip for the `map` function is visible, showing its signature `map(f, c...) -> collection` and a description: "Transform collection `c` by applying `f` to each element. For multiple collection arguments, apply `f` elementwise." It also includes an example: `julia> map(x -> x * 2, [1, 2, 3])` resulting in `3-element Array{Int64,1}: 2`.

The `Julia Plots (2/2)` panel on the right shows a plot of a sine wave. The x-axis ranges from 4 to 8, and the y-axis ranges from 0.0 to 1.0. The plot is labeled `y1`.

The bottom panel shows the `TERMINAL` output, which displays the output of the Julia script, including the iteration numbers 4 through 10. The status bar at the bottom indicates the current file is `sp/isdeflocals*`, the environment is `Julia env: v1`, and the cursor is at `Ln 17, Col 4`.

Julia on bwUniCluster 3.0

Access bwUniCluster 3.0 via SSH

Terminal

```
ssh <yourlogin>@uc3.scc.kit.edu
```

VS Code

Remote-SSH: Connect to Host...

```
<yourlogin>@uc3-login1.scc.kit.edu
```

```
<yourlogin>@uc3-login2.scc.kit.edu
```


Loading software modules

For this course

```
module load juliahpc
```

This loads Julia, CUDA and OpenMPI.

Outside of the course: If there is no (working) system module, use standard binaries provided by [juliaup](#).

Jobs are scheduled with SLURM.

Submit a job:

```
sbatch job-script.sh
```

Check on your queued/running jobs:

```
squeue --me
```


Side remark: Accessing compute nodes with VS Code

On the target node

```
module load code  
code tunnel
```

On your laptop

In VS Code:

```
Remote Tunnels: Connect to Tunnel
```

Let's try it!

→ `cd JuliaKIT26`

→ `git pull`

→ `jupyter lab`

→ `exercises/Day1/1_onboarding`

